

Sanket Kumar

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SUMMARY

Computer Science undergraduate specializing in Data Science with hands-on experience in Backend Development, DevOps, and Artificial Intelligence through academic projects. Skilled in Java, Python, REST APIs, databases, containerization, and CI/CD, with experience building full-stack and automation-driven systems.

EDUCATION

Vellore Institute of Technology <i>Bachelor of Technology in Computer Science CGPA: 8.69</i>	2023 – Present Vellore
Baldwin Farm Area High School <i>Class XII 80.8%</i>	2023 Jamshedpur
Baldwin Farm Area High School <i>Class X 93%</i>	2021 Jamshedpur

PROJECTS

- PulseOS – Healthcare Interoperability Platform** | *React, TypeScript, FastAPI, PostgreSQL, FHIR R4*
- Engineered a full-stack healthcare interoperability platform connecting patient, doctor, and hospital workflows using FHIR R4, FastAPI, React/TypeScript, and PostgreSQL.
 - Implemented patient-controlled consent and role-based access control with Firebase authentication, Health ID lookup, record visibility controls, and Redis rate limiting.
 - Integrated Synthea-generated FHIR data with PostgreSQL and MinIO, and built automated tests for registration, authorization, audit logging, and CORS.
- DevOps-Enabled To-Do REST API** | *Java, Jenkins, Docker, Terraform, Ansible*
- Developed and tested a RESTful Java API using JUnit and implemented an automated CI/CD pipeline with Jenkins.
 - Containerized the application using Docker and automated infrastructure provisioning with Terraform and Ansible.
 - Implemented application and system monitoring using Graphite and Grafana dashboards to track performance and health.
- Adaptive Cyber Defense System** | *Python, PyTorch, NumPy, Reinforcement Learning*
- Developed a multi-adversary cyber defense system using Deep Q-Networks (DQN) for adaptive security decision-making.
 - Simulated evolving attacker behavior and dynamic defense actions including Allow, Block, Throttle, Observe, and Deceive.
 - Trained the reinforcement learning model for 30,000 simulation steps and achieved stable policy convergence.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: React.js, FastAPI, Node.js, TensorFlow, TensorFlow Lite, PyTorch, OpenCV, NumPy, JUnit
Databases & Storage: PostgreSQL, Oracle SQL, MySQL, Firebase, MinIO
Cloud & DevOps: Docker, Jenkins, Terraform, Ansible, Git, GitHub, Redis, Graphite, Grafana
Other: FHIR R4, Synthea, Raspberry Pi, Firebase Authentication, RBAC
Concepts: Data Structures & Algorithms, Object-Oriented Programming, Compiler Design, Operating Systems, REST API Development, Artificial Intelligence, DevOps, Cloud Computing

CERTIFICATIONS

- DevOps, Agile & Design Thinking** – IBM Career Education Program *Jul 2026*
Credential ID: CEDEA11N — Certificate
Covered DevOps practices, Agile workflows, Design Thinking, and CI/CD pipeline fundamentals.
- DevOps Fundamentals** – IBM Career Education Program *Jun 2025*
Credential ID: CEDEV2IN — Certificate
Covered Docker, automation, cloud deployment, infrastructure provisioning, and CI/CD fundamentals.

LEADERSHIP & EXTRACURRICULARS

- Senior Core Member, Geospatial Club – VIT** *Technical Domain*
Developed and maintained the club website while organizing technical workshops and events.